Analog and Digital Electronics_2024_PYQ

Group-A (Very Short Answer Type Question)

Answer any ten of the following: (1 x 10 = 10)

1.

a) What will be the minimal product-of-sums function described by the K-map given in Fig.?

C \ AB	00	01	11	10
0	1	1	ϕ	0
1	0	0	ϕ	0

b) Arrange the Classes of power amplifier according to their efficiency (low to high): Class A, Class B, Class C, Class AB

c) What will be the output of the logic gate in the figure given below?

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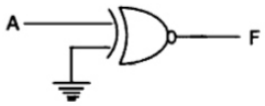
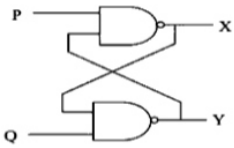
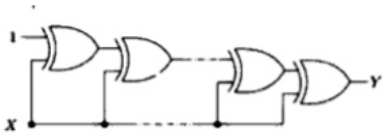


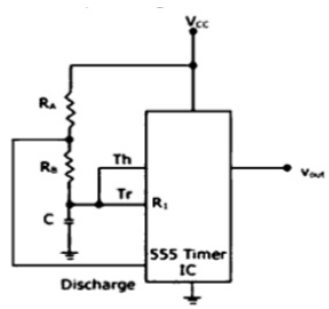
d) In the latch circuit shown, the NAND gates have non-zero, but unequal, propagation delays. The present input condition is: P = Q = "0". If the input condition is changed simultaneously to P = Q = "1", what will be the outputs X and Y now?



e) If the input to the digital circuit (in the figure) consisting of a cascade of 20 XOR-gates is X. What will be the output Y ?

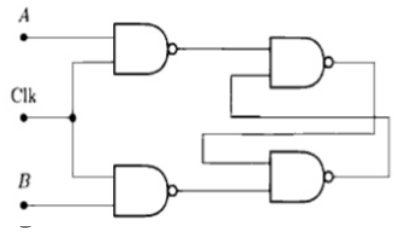


f) The circuit of following figure shows an IC 555 Timer connected as an Astable multivibrator. The value of the capacitor C is 10 nF. Find the values of the resistors R_A and R_B for a frequency of 10 kHz and a duty cycle of 0.75 for the output voltage.



g) What will be the minimum number of 2 to 1 multiplexer required to realize a 4 to 1 multiplexer?

h) Consider the given circuit. Explain whether the race around will occur or not.



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i) Which A/D converter is used for Hum rejection?

j) The A.C. output power of a Class B push-pull power amplifier is 10 watts. What will be the D.C. input power drawn from power supply when the efficiency is maximum?

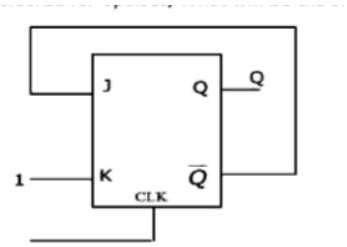
k) What is octal value of $(2^6)_{10}$?

l) Statement 1: A stable Multivibrator can be used for generating Square Wave.
Statement 2: Bi stable Multivibrator can be used for storing binary information.
Please state whether both the statements are true or false.

Group-B (Short Answer Type Question)

Answer any three of the following: (5 x 3 = 15)

2. In a JK flip-flop, we have J = Q' and K = 1 (see figure). Assuming the flip-flop was initially cleared



and then clocked for 6 pulses, what will be the sequence at the Q output?

3. Derive the maximum efficiency of a class B amplifier.

4. Explain the operation of transformer coupled Class A amplifier.

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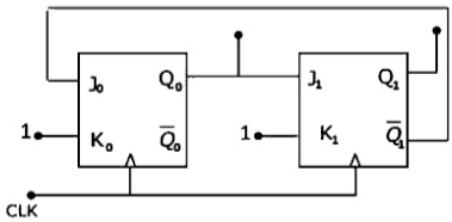

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5. Find out what will be the modulus of this counter? (Counter circuit diagram).

6. Design a Full subtractor (X, Y and Borrow) with 4:1 MUX.

Group-C (Long Answer Type Question)

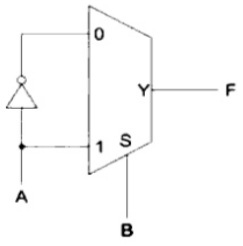
Answer any three of the following: (15 x 3 = 45)

7.

(a) Find the out expression for the Karnaugh Map shown below:

AB \ CD	00	01	11	10
00	0	0	0	0
01	1	0	0	1
11	1	0	1	1
10	0	0	0	0

(b) Find the out F in terms of A and B (Figure: MUX circuit).



(c) Find the output expression F for the following Karnaugh Map and realise it with logic gates:

C \ AB	00	01	11	10
0	1	1		
1		1	1	

8.

(a) A digital circuit compares two numbers $A_3A_2A_1A_0$(A) and $B_3B_2B_1B_0$(B). Find the pair A, B to get output Y = 0.

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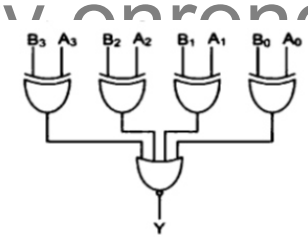
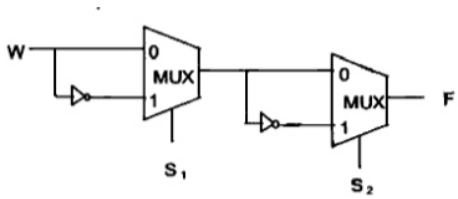


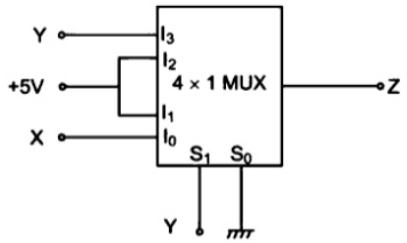
(b) Consider the multiplexer-based logic circuit shown. Find the Boolean function realized by the circuit.



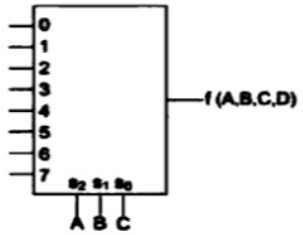
(c) A 4-bit modulo-16 ripple counter uses JK flip-flops. If the propagation delay of each FF is 50ns, find the maximum clock frequency that can be used in this counter.

9.

(a) Find the output Z in terms of X and Y for a 4 x 1 MUX with specified connections.



(b) A Boolean function F(A, B, C, D) = π(1, 5, 12, 15) is to be implemented using an 8x1 multiplexer (A is MSB). The inputs ABC are connected to the select inputs S2S1S0 of the multiplexer.



Find the correct inputs to pins 0, 1, 2, 3, 4, 5, 6, 7 in order for the shown 8:1 multiplexer (function of A, B, C, D).

10.

(a) A 3-bit gray counter is used to control the output of the multiplexer as shown in the figure (A2 is MSB, A0 is LSB). The initial state of the counter is 000_2. The output is pulled high. Find the sequence of the output of the circuit.

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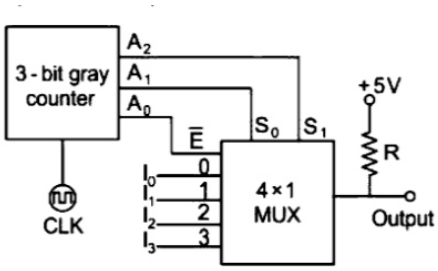
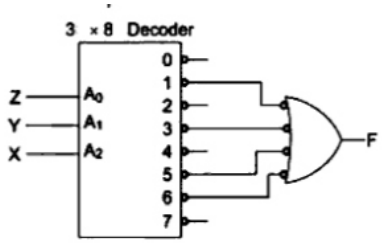


(b) A 3 line to 8-line decoder, with active low outputs, is used to implement a 3-variable Boolean function as shown in the figure.

Find the simplified form of Boolean function F(A, B, C) implemented in 'Product of Sum' form.



11.

(a) Digital input signals A, B, C with A as the MSB and C as the LSB are used to realize the Boolean function F = m0 + m2 + m3 + m5 + m7, where mi denotes the ith m term. In addition, F has a don't care for m1. Find the simplified expression for F.

(b) Find the prime implicants in the sum of products function (X, Y, Z) = Σ(2, 3, 4, 5).

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